

Xenohormesis: health benefits from an eon of plant stress response evolution

Xenohormesis is a biological principle that explains how environmentally stressed plants produce bioactive compounds that can confer stress resistance and survival benefits to animals that consume them. Animals can piggyback off products of plants' sophisticated stress response which has evolved as a result of their stationary lifestyle. Factors eliciting the plant stress response can judiciously be employed to maximize yield of health-promoting plant compounds. The xenohormetic plant compounds can, when ingested, improve longevity and fitness by activating the animal's cellular stress response and can be applied in drug discovery, drug production, and nutritional enhancement of diet.